Predictive Insight Engine

Theory Concepts & Definitions (supporting document)

1. Supervised Learning

Supervised learning learns a function f: X -> y from **labelled** examples (inputs with known outputs) and then predicts y for new inputs. The two main problem types are regression and classification.

2. Regression vs Classification

- **Regression** predicts a continuous number (house price, temperature); typical metrics: MSE, RMSE, MAE, R2.
- **Classification** predicts a discrete category (spam / not spam); typical metrics: accuracy, precision, recall, F1.

3. Simple Linear Regression (SLR)

- Model: **y = b0 + b1*x + e**. b0 = intercept (value of y when x = 0), b1 = slope (change in y for one unit increase in x), e = random error.
- Parameters are estimated by Ordinary Least Squares (OLS): minimise the sum of squared residuals. b1 = Cov(x,y)/Var(x), b0 = mean(y) - b1*mean(x).

4. Multiple Linear Regression (MLR)

Model: **y = b0 + b1*x1 + b2*x2 + ... + bp*xp + e**. Each coefficient is the change in y for a one-unit change in that feature *holding the others constant*. Closed form (Normal Equation): w = (X'X)^-1 X'y.

5. Polynomial Regression

Linear regression applied to polynomial features (x, x^2, x^3, ... and interaction terms). It is still *linear in the parameters* but can fit curves. Higher degree = more flexibility = higher risk of overfitting.

6. Assumptions of Linear Regression

- **Linearity** - relationship between X and y is linear (check: scatter / residual-vs-fitted plot).
- **Independence** of errors - no autocorrelation (Durbin-Watson ~ 2).
- **Homoscedasticity** - constant variance of residuals (no funnel shape).
- **Normality** of residuals (histogram, Q-Q plot).
- **No multicollinearity** among predictors (VIF < 5-10).

7. Evaluation Metrics

- **MSE** = (1/n) sum (y - y_hat)^2 - penalises large errors, units squared.
- **RMSE** = sqrt(MSE) - error in the units of y.
- **MAE** = (1/n) sum |y - y_hat| - average absolute error, robust to outliers.
- **R2** = 1 - SS_res/SS_tot - share of variance explained (1 = perfect, 0 = predicts the mean).

- **Adjusted R2** = 1 - (1 - R2)(n - 1)/(n - p - 1) - R2 penalised for the number of predictors p.

8. Train/Test Split & Cross-Validation

The data is split (here 80/20) so the model is judged on unseen data. k-fold cross-validation repeats training/validation on k different folds to give a more stable estimate of generalisation error.

9. Overfitting and Underfitting

- **Underfitting**: model too simple; high error on both training and test data (high bias).
- **Overfitting**: model too complex; very low training error but much higher test error (high variance); it memorises noise.

10. Bias-Variance Trade-Off

- Expected prediction error = **Bias^2 + Variance + Irreducible noise**.
- **Bias** = error from wrong/too-simple assumptions. **Variance** = sensitivity to the particular training sample.
- Increasing complexity lowers bias but raises variance; the best model sits at the minimum of the total error (U-shaped test-error curve).

11. Gradient Descent

- Iterative optimiser: w := w - alpha * grad J(w), where J is the cost (MSE/2) and alpha the learning rate. For linear regression J is convex, so GD reaches the global minimum.
- **Batch GD** - uses all samples per update (stable, slow per step). **SGD** - one sample per update (fast, noisy). **Mini-Batch GD** - small batches (good compromise, most used).
- Feature scaling (standardisation) is required for fast, stable convergence; too large alpha diverges, too small alpha is slow.

12. Feature Scaling, Multicollinearity, VIF

- **Standardisation**: z = (x - mean)/std, computed on training data only (avoids data leakage).
- **Multicollinearity**: predictors strongly correlated with each other; makes individual coefficients unstable. **VIF** = 1/(1 - R2_j), where R2_j comes from regressing feature j on the others.

Quick comparison of the regression models used

Model	Features	Strength	Weakness
Simple Linear	1	Easy to interpret	High bias if one feature is not enough
Multiple Linear	p	Interpretable, stable, uses more information	Cannot model curvature / interactions
Polynomial	p -> many terms	Captures non-linear effects and interactions	Variance grows quickly with degree